

For large diameters, G can be approximated with that for an infinite plane sample (section D.1). The influence of the finite diameter is the greater, the smaller t . Therefore we have the upper limit for the influence of the periferi in the correction factor C_0 of section I.1 for a thin, circular slice, and we can write

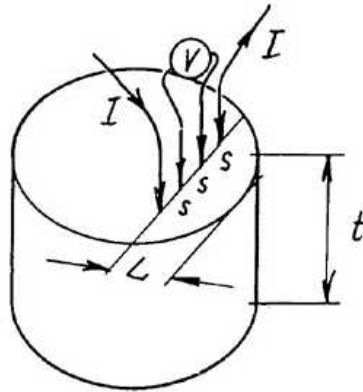
$$\rho = G \frac{V}{I}, \quad 2\pi s \cdot T_1\left(\frac{t}{s}\right) \cdot C_0\left(\frac{d}{s}\right) \leq G \leq 2\pi s \cdot T_1\left(\frac{t}{s}\right)$$

where:

$2\pi s \cdot T_1\left(\frac{t}{s}\right)$ is the geometric factor for an infinite plane sample of thickness t , and $C_0\left(\frac{d}{s}\right)$ is the diameter correction for a thin circular slice of diameter d , when measuring in the center.

$T_1\left(\frac{t}{s}\right)$ is found in section D.1, and $C_0\left(\frac{d}{s}\right)$ in section I.1.

H.3) Probe Array Perpendicular to a Diameter at a fixed Distance from the Periferi.



This configuration has not been treated in the literature.

By a reasoning analogous to that in the previous section H.2, we conclude that

$$\rho = G \frac{V}{I}, \quad \text{where } 2\pi s \cdot T_1\left(\frac{t}{s}\right) K_3\left(\frac{L}{s}, \frac{d}{s}\right) \leq G \leq 2\pi s \cdot D_T\left(\frac{L}{s}, \frac{t}{s}\right)$$

where:

$2\pi s \cdot D_T\left(\frac{L}{s}, \frac{t}{s}\right)$ is the geometric factor for a semi-infinite plane sample of thickness t , when the probe array is parallel to the edge at a distance L , (see section E.4), and $2\pi s \cdot T_1\left(\frac{t}{s}\right)$ is the geometric factor for an infinite plane sample of thickness t (see section D.), and $K_3\left(\frac{L}{s}, \frac{t}{s}\right)$ is the contour correction for the shown configuration, when $t \ll s$ (see section I.4).